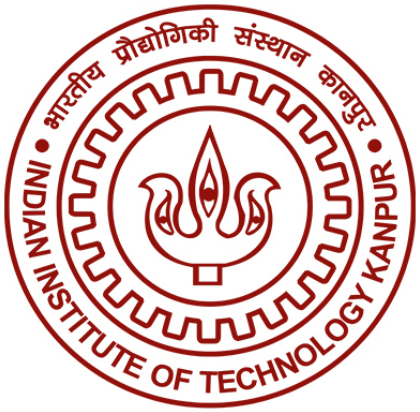


SPA617: RADIO ASTRONOMY LAB

Antenna Characterization

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Lab Report

1 AIM

The aim of this experiment is to characterize the radial directional and polarization-dependent response of a horn antenna using a dipole antenna as the transmitting source.

2 Objectives and Learning Outcomes

The objectives of this experiment are:

1. To understand the fundamental operating principles and construction of dipole and horn antennas.
2. To study the variation of the received signal intensity with the distance between the transmitting dipole and receiving horn antenna.
3. To experimentally determine the directional characteristics of the horn antenna by measuring the received signal as a function of its angular orientation.
4. To measure and compare the antenna response in the E-plane and H-plane configurations.
5. To examine the influence of relative polarization between the transmitting and receiving antennas on the received signal strength.
6. To characterize the horn antenna beam pattern and determine parameters such as the direction of maximum response and beamwidth.
7. To investigate the effects of reflections, scattering, and other environmental factors on practical antenna measurements.

3 Theory

As an electromagnetic wave propagates through free space, its power density decreases with increasing distance from the transmitting antenna. In the far-field region, the received power approximately follows the inverse-square law,

$$P_r \propto \frac{1}{r^2},$$

where P_r is the received power and r is the separation between the transmitting and receiving antennas. Polarization describes the orientation of the electric-field vector of an electromagnetic wave. For a linearly polarized wave, the electric field oscillates in a fixed direction perpendicular to the direction of propagation. Maximum received signal is obtained when the receiving antenna is aligned with the polarization of the incident wave, whereas a relative rotation reduces the received signal strength. In this experiment, co-polarized and cross-polarized configurations were studied. In the co-polarized configuration, the transmitting and receiving antennas have aligned polarization, while in the cross-polarized configuration, they are oriented perpendicular to each other. The directional response of the horn antenna was measured in the E-plane and H-plane, defined according to the orientation of its sensing element. Rotating the

antenna in these planes enabled the angular variation of the received signal to be measured and the corresponding antenna beam patterns to be characterized.

Directivity quantifies the ability of an antenna to concentrate its radiated power in the direction of maximum radiation compared with an isotropic radiator. It is related to the beam solid angle, Ω_A, by

$$D = \frac{4\pi}{\Omega_A}. \tag{1}$$

For a practical antenna, the directivity can be approximated using the half-power beamwidths in the E-plane and H-plane as

$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{2}$$

where θ_E and θ_H are the half-power beamwidths in the E-plane and H-plane, respectively, expressed in radians.

The beam pattern of an antenna represents the variation of its radiated or received power with respect to direction. It typically consists of a main lobe, corresponding to the direction of maximum radiation, along with possible side lobes and a back lobe. Experimentally, the beam pattern can be determined by rotating the receiving antenna through different angles and recording the corresponding received power.

4 Equipments

The apparatus used for the antenna characterization experiment are:

- Horn antenna
- Dipole antenna
- Signal generator
- Spectrum analyzer
- Measuring tape with a least count of 1 mm
- Angular measurement scale

5 Experimental setup

The experimental arrangement comprised a dipole antenna driven by a signal generator and a horn antenna connected to a spectrum analyzer. To study the dependence of received power on distance, the separation d between the two antennas was varied while keeping their orientation unchanged. For the beam-pattern measurements, the antenna separation was kept constant, and the horn antenna was rotated about its vertical axis by an angle θ. The received power was then recorded at each distance and angular position using the spectrum analyzer.

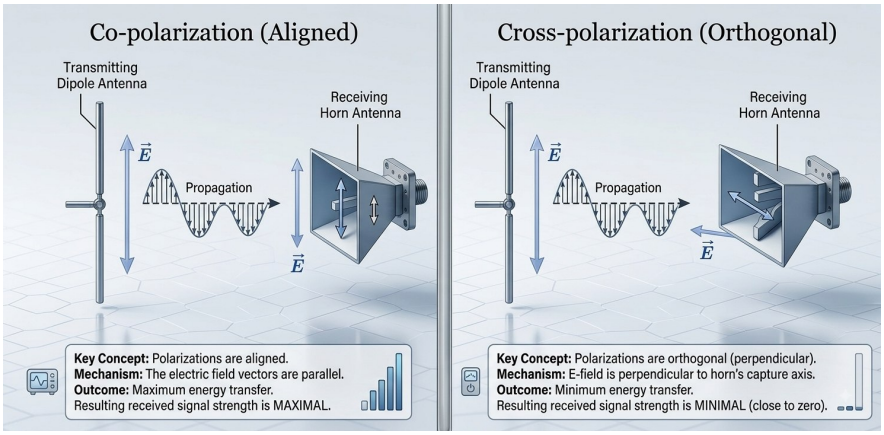


Figure 1: Experimental setup for Co and Cross polarization configuration.

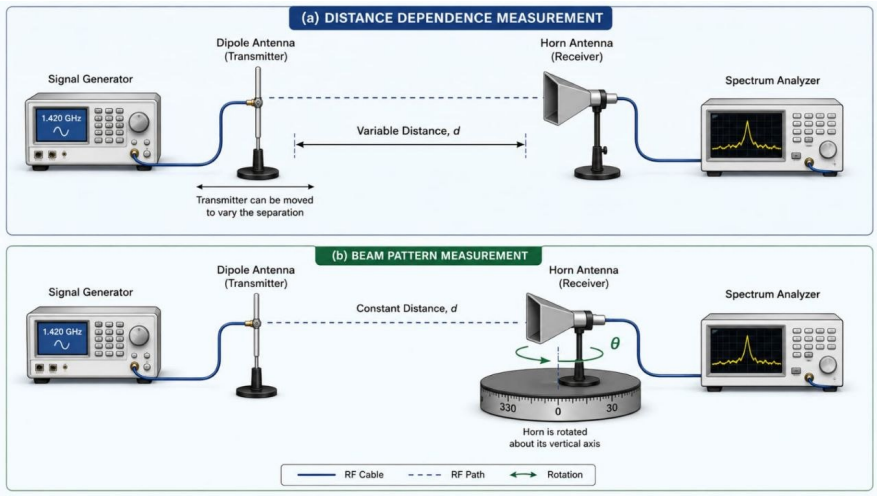


Figure 2: Experimental setup for distance-dependance and beaam-pattern measurements.

6 Observation

The experiment was conducted under constant operating conditions for both the transmitting and receiving instruments. The signal generator was set to a frequency of 1.42 GHz and an output level of 20. The spectrum analyzer was configured with a reference level of -20 and an attenuation of 20 dB, with the received signal measured in dBm. These frequency and instrument settings were kept unchanged throughout the experiment to ensure that variations in the measured received power were primarily due to changes in the separation distance or angular orientation. The recorded received-power values for the different experimental configurations are presented in the table below. For the distance-dependence measurements, the separation between the transmitting dipole and receiving horn was varied while maintaining constant instrument settings.

Table 1: Power recieved vs angle (Co-polarization)

Angle	**Power(dBm) for E plane**	**Power(dBm) for H plane**
0	-36.40	-45.24
30	-33.27	-48.37
60	-37.80	-53.92
90	-46.20	-62.62
120	-45.01	-55.09
150	-42.97	-51.50
180	-37.32	-52.32
210	-41.96	-43.93
240	-50.20	-48.85
270	-49.26	-57.14
300	-39.43	-52.72
330	-33.42	-45.95
360	-32.00	-45.22

Table 2: Power received vs angle (Cross-polarization)

Angle	**Power(dBm) for E plane**	**Power(dBm) for H plane**
0	-49.54	-41.47
30	-47.50	-43.32
60	-50.77	-40.40
90	-60.11	-43.09
120	-59.44	-42.33
150	-54.42	-47.71
180	-59.51	-41.73
210	-54.90	-47.40
240	-58.32	-59.33
270	-65.46	-49.34
300	-54.40	-44.29
330	-48.93	-40.96
360	-50.84	-40.97

Table 3: Distance vs power recieved table

Length	**Power (dBm)**	**Received Power (dBm)**	$20log_{10}(Length)$
217	20	-28.80	46.73
343	20	-26.50	50.71
418	20	-37.60	52.42
495	20	-34.55	53.89
572	20	-40.57	55.15
694	20	-46.90	56.83
726	20	-28.70	57.22
803	20	-33.63	58.09

7 Results

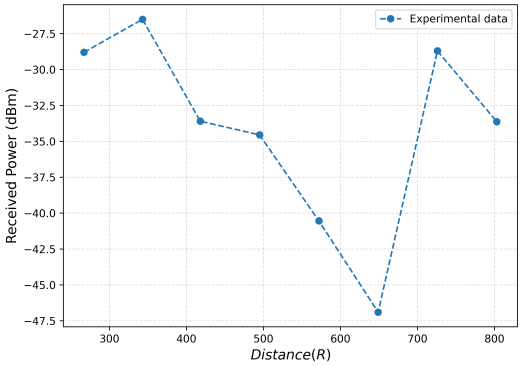


Figure 3: Power output as function of distance

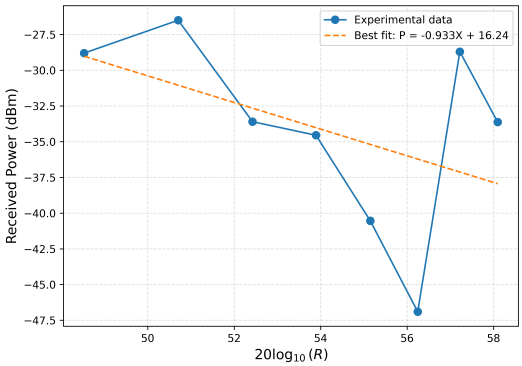


Figure 4: Power output as a function of $20log_{10}(Length)$

In the above mentioned plots the dependence of power received with the change in distance has been recorded. The noticeable variation of power outputs are expected with the $\frac{1}{R^2}$ dependence. But it seems to violate at 726 cm, actually this was the point we reached the walls of the room that's why the reflected waves from the walls itself gave a push in the power output.

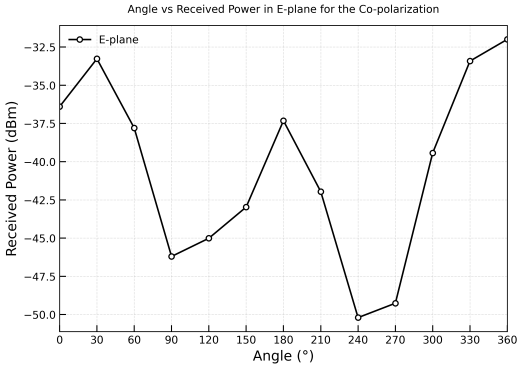


Figure 5: E- plane co-polarization

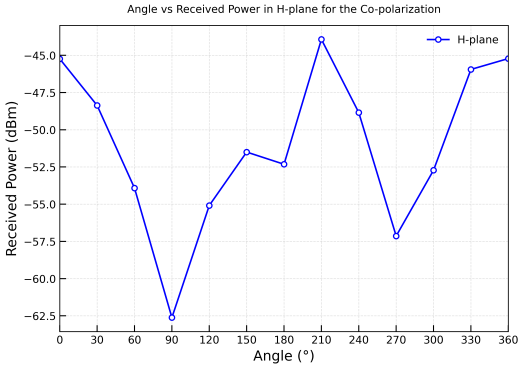


Figure 6: H-Plane Co-polarization

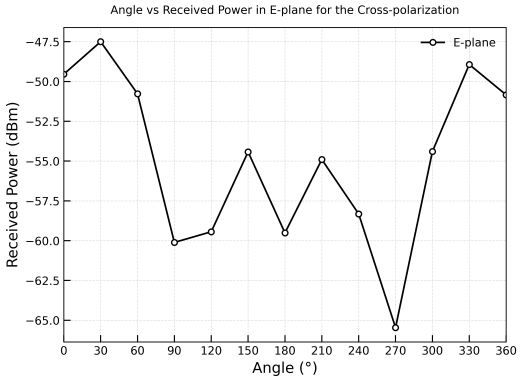


Figure 7: E- plane cross-polarization

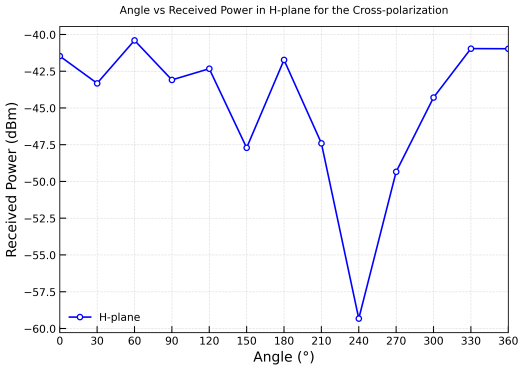


Figure 8: H-Plane Cross-polarization

When measuring the received power, the receiving antenna's orientation in both the E and H planes made a clear difference. In the co-polarized setup, peak power reached roughly $-32.00\ dBm$ in the E-plane and slightly higher at $-43.93\ dBm$ in the H-plane. The cross-polarized setup naturally yielded weaker signals, though its peak levels and angular variations still differed between the two planes.

Across all four angular patterns, we noticed multiple peaks rather than a single clean response. This shows that the antenna picks up significant signal levels over a broad range of angles. These extra peaks are likely caused by multiple effects like reflections, scattering, and interference from surrounding objects in the lab rather than the antenna's pure radiation pattern.

The contrast between polarization states was plain to see. Co-polarized alignment delivered stronger overall signals because the incoming wave aligned well with the receiving antenna. Cross-polarization reduced the overall power, but its pattern still showed distinct peaks. On top of that, the E and H plane responses were quite different, confirming that the plane of rotation directly shapes how the antenna responds.

Ultimately, these results confirm that both orientation and polarization alignment play a major role in signal reception. At the same time, the extra peaks and pattern distortions show just how much the surrounding lab environment can influence real-world antenna measurements.

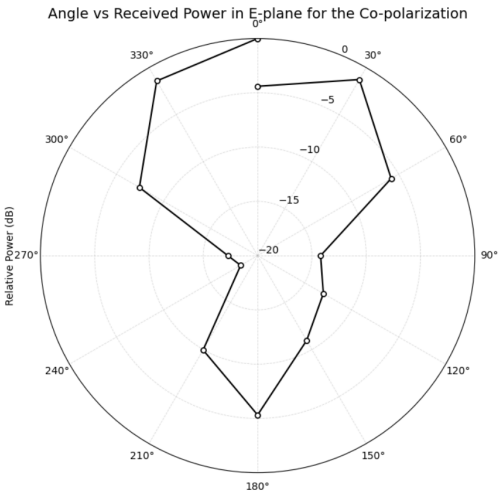


Figure 9: E-plane Co-polarization

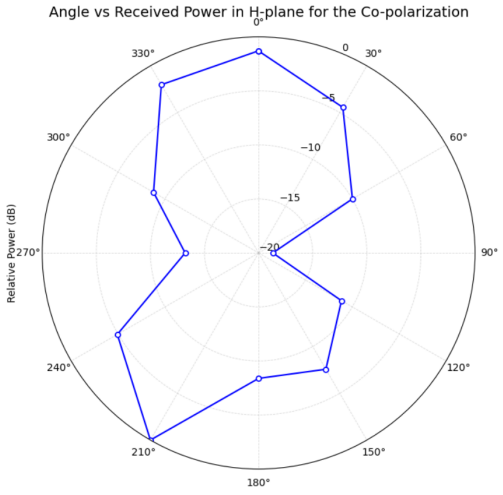


Figure 10: H-plane Co-polarization

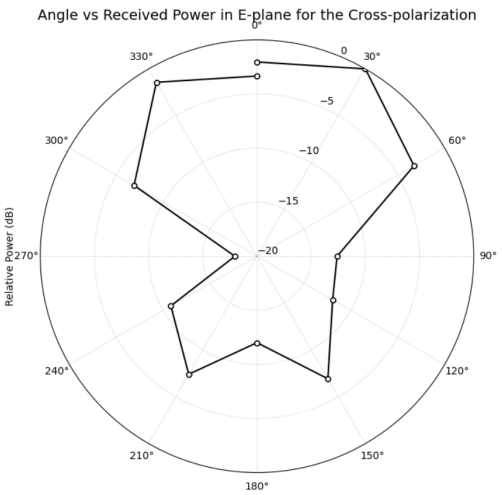


Figure 11: E-plane Cross-polarization

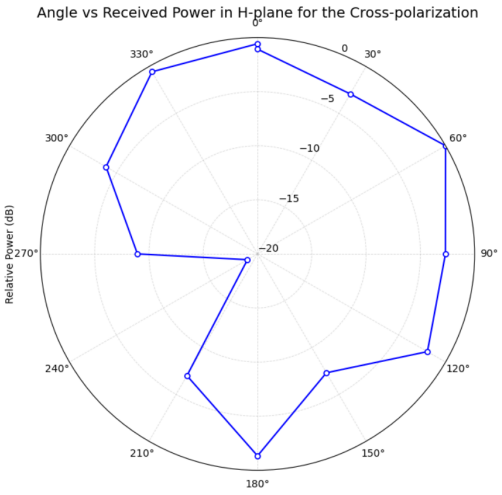


Figure 12: H-plane Cross-polarization

In our polar plots we normalized the power to the maximum value for clearer plot. The normalized beam patterns for both the E and H-planes make it clear that the antenna isn't strictly unidirectional. Instead of focusing power into a single, tight main lobe, all four configurations picked up noticeable signals across a wide range of angles.

We also saw several prominent side lobes and back lobes. In the co-polarized patterns, the response remained surprisingly strong even when facing away from the primary direction, especially on the back side of the antenna. The cross-polarized setups showed similar secondary features, proving that signal reception doesn't drop to zero when the antenna is turned off-axis. While some of these extra lobes are inherent to the antenna design itself, lab reflections off nearby walls and equipment definitely played a role, creating interference that reshaped the measured pattern.

8 Calculation

Operating Frequency Calculations

The pole length of the horn antenna was measured to be

$$l = 4.5 \text{ cm}.$$

Assuming

$$\frac{\lambda}{4} = 4.5 \text{ cm},$$

we obtain

$$\lambda = 18 \text{ cm} = 0.18 \text{ m}.$$

Therefore,

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{0.18} \approx 1.67 \text{ GHz}.$$

The actual operating frequency was approximately

$$\boxed{f_{\text{actual}} = 1.42 \text{ GHz}}.$$

Far-Field Calculations

The far-field distance is given by

$$R_{\text{FF}} = \max\left(3\lambda, \frac{2d^2}{\lambda}\right).$$

For the horn antenna,

$$d = 35 \text{ cm}, \qquad \lambda = 12.13 \text{ cm}.$$

Hence,

$$R_{\text{FF}} = \max\left(3(12.13), \frac{2(35)^2}{12.13}\right).$$

Thus,

$$R_{\text{FF}} = \max(36.39, 201.98) = 201.98 \text{ cm}.$$

Therefore,

$$\boxed{R_{\text{FF}} \approx 2.02 \text{ m}}.$$

HPBW and Directivity

The maximum received power in the E-plane was measured to be

$$P_{\max,E} = -32.00 \text{ dBm}.$$

The half-power level corresponds to a decrease of approximately 3 dB from the maximum received power. Therefore,

$$P_{-3\text{dB},E} = P_{\max,E} - 3 = -32.00 - 3 = -35.00 \text{ dBm}.$$

The first -3 dB crossing lies between 0°and 30° as -36.40 dBm and -33.27. Using linear interpolation,

$$\theta_{E,L} = 0 + \frac{30 - 0}{-35 + 36.40}\left[-33.27 + 36.40\right],$$

giving

$$\theta_{E,L} \approx 13.42^\circ.$$

The second -3 dB crossing lies between 300°and 330° as -39.43 dBm and -33.43. Using linear interpolation,

$$\theta_{E,R} = 300 + \frac{330 - 300}{-35 - (-39.43)}\left[-33.43 + 39.43\right],$$

giving

$$\theta_{E,R} \approx 37.85^\circ.$$

Hence, the E-plane half-power beamwidth is

$$\text{HPBW}_E = \theta_{E,R} - \theta_{E,L},$$

and therefore,

$$\boxed{\text{HPBW}_E \approx 24.43^\circ}.$$

For the H-plane, the maximum received power was

$$P_{\max,H} = -43.93 \text{ dBm}.$$

Therefore, the corresponding -3 dB level is

$$P_{-3\text{dB},H} = P_{\max,H} - 3 = -43.93 - 3 = -46.93 \text{ dBm}.$$

The first -3 dB crossing lies between 0° and 30° as $-45.24\ dBm$ and -48.37. Linear interpolation gives

$$\theta_{H,L} = 0 + \frac{30 - 0}{-46.93 + 45.24}\left[-48.37 + 45.24\right],$$

and hence,

$$\theta_{H,L} \approx 16.20^\circ.$$

The second -3 dB crossing lies between 300° and 330° as $-52.72\ dBm$ and -45.95. Thus,

$$\theta_{H,R} = 300 + \frac{300 - 330}{-46.93 + 52.72}\left[-45.95 + 52.72\right],$$

giving

$$\theta_{H,R} \approx 34.34^\circ.$$

Therefore, the H-plane half-power beamwidth is

$$\text{HPBW}_H = \theta_{H,R} - \theta_{H,L},$$

giving

$$\boxed{\text{HPBW}_H \approx 18.14^\circ}.$$

The beamwidths are converted to radians before calculating the directivity:

$$\theta_E = 24.43^\circ \left(\frac{\pi}{180}\right) \approx 0.0.43 \text{ rad},$$

and

$$\theta_H = 34.34^\circ \left(\frac{\pi}{180}\right) \approx 0.32 \text{ rad}.$$

Using the beamwidth approximation for directivity,

$$D \approx \frac{4\pi}{\theta_E \theta_H},$$

we obtain

$$D \approx \frac{4\pi}{(0.43)(0.32)} \approx 91.32.$$

Therefore, the estimated directivity of the antenna is

$$\boxed{D \approx 91.32}.$$

VSWR, Reflection Coefficient and return loss

The minimum VSWR is given by

$$\text{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1.75}{0.7} = 2.5.$$

The reflection coefficient is

$$|\Gamma| = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{2.5 - 1}{2.5 + 1} \approx 0.43.$$

The return loss is given by

$$\mathrm{RL} = -20\log_{10}|\Gamma| = -20\log_{10}(0.43) \approx -12.67 \text{ dB}.$$

Therefore,

$$\boxed{\mathrm{VSWR} = 2.5}$$

$$\boxed{|\Gamma| \approx 0.43}$$

$$\boxed{\mathrm{RL} \approx -12.67 \text{ dB}}$$

9 Conclusion

This hands-on experiment gave us a much clearer picture of how antennas operate and how to extract key properties like directivity straight from a beam pattern. Rather than focusing energy in just one direction, the antenna picked up strong signals across multiple angles. Crucially, the test also highlighted the impact of the surrounding environment showing firsthand how real-world factors like reflections and wave interference can alter measured antenna performance. How measuring beam pattern closer than far-field distance cause the wavefronts arriving from different parts of antenna become strongly curved rather than effectively planar which distorts the output. Overall, it was a practical, insightful way to connect electromagnetic theory with actual radio-wave behavior.

Source of uncertainty

- Environmental & Reflections: Signals bouncing off lab walls, tables, and equipment create multipath interference, altering the measured power.
- Manual Rotation Errors: Slight inaccuracies when manually aligning and turning the antenna introduce uncertainty into the beam pattern.
- Polarization & Axis Misalignment: Any slight offset between the transmitting and receiving antennas directly impacts the precision of co-pol and cross-pol readings.
- Instrument Noise & Signal Drift: Noise in the equipment and minor signal fluctuations cause variations, especially when measuring lower power levels.